

ARGUS: Auditing Causal Identification Credibility in Environmental Policy Evaluation

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Abstract. Empirical economics leans heavily on difference-in-differences (DID) designs—nowhere more visibly than in environmental policy, our motivating case—whose credibility rests on identification assumptions that are often asserted rather than audited. We propose **ARGUS**, a responsible-AI framework that audits the *identification credibility* of such studies rather than adjudicating the true causal effect, which is never observed. ARGUS decomposes identification into auditable dimensions, gathers in-paper evidence, assesses each assumption–implication–evidence chain, and localizes risk in a transparent, human-reviewable report. We validate the auditor with *synthetic flaw injection*, which provides reproducible local ground truth without the true effect. Once the leakage that makes naive evaluation circular is removed, a deterministic keyword baseline detects only 0.182 of injected flaws and is blind to flawed-but-present evidence; swapping only the assessment policy to a model that judges evidence *adequacy* recovers detection to 1.000 with no false alarms, showing that evidence presence is not evidence adequacy. On 27 real published DID papers the binding constraint shifts to *evidence grounding*: the framework honestly reports retrieval failure rather than over-flagging, localizing where real-paper auditing is hard.

Keywords: Responsible AI · Causal inference · Difference-in-differences
· Environmental policy · LLM agents.

1 Introduction

Difference-in-differences (DID) is a workhorse design across empirical economics, and a particularly visible one in environmental policy—our motivating case. Studies of carbon pricing, emissions-trading pilots, and industrial-emission standards routinely compare treated and untreated units before and after a policy to recover its causal effect. The credibility of any such estimate rests entirely on a chain of *identification assumptions*: that treated and control units would have moved in parallel absent the policy, that units did not anticipate it, that staggered adoption is handled with an appropriate estimator, that there is no interference between treated and control units, and so on. A decade of methodological work has shown how fragile each of these assumptions can be, and how easily a standard two-way fixed-effects specification can mislead under heterogeneous and staggered treatment timing [4,1,10,2,9].

Yet in applied work these assumptions are frequently *asserted* rather than *audited*. A paper states that parallel trends “hold”, shows an event-study plot, and moves on; a reader who wants to know whether the identification argument is actually supported by the reported evidence must reconstruct it by hand, dimension by dimension. This is slow, expert-intensive, and inconsistent across reviewers. The result is a reliability gap in a body of evidence that increasingly informs environmental policy.

Our stance. We argue that identification *credibility* is auditable even though the causal *effect* is not. In a DID design the counterfactual is never observed, so it is hopeless—and, we contend, the wrong target—to ask an automated system “is the estimated effect true?”. Instead we ask a question that can be grounded in the paper itself: *is each identification assumption supported by the evidence the study reports, and where is that support weakest?* This reframing turns an unanswerable causal question into a tractable auditing problem over assumptions, their testable implications, and the in-paper evidence that bears on them. Figure 1 illustrates the view: credibility varies across assumptions, evidence, and policy context, and a useful system should localize where it is fragile rather than issue a verdict.

ARGUS. We introduce **ARGUS**, a responsible-AI framework that audits the identification credibility of DID environmental-policy studies. ARGUS decomposes identification into a fixed set of auditable dimensions, gathers the supporting evidence each dimension requires, assesses every assumption–implication–evidence chain, localizes risk across dimensions, and produces a transparent, evidence-anchored report for a human expert. It is deliberately a *bounded-agentic* system: the pipeline between modules is fixed and deterministic, and model-driven agency is confined to two scoped sub-tasks. Because the true effect is unavailable as ground truth, we validate the auditor with *synthetic flaw injection*: we perturb a credible paper with one known identification flaw and check whether ARGUS detects it, stays quiet on the clean version, and localizes the right dimension. The injected flaw is local ground truth that requires no access to the true effect.

Contributions. This paper, presented as a doctoral-research plan, makes four contributions. (1) **ARGUS**, a bounded-agentic framework that audits identification credibility—not causal truth—in DID environmental-policy evaluation (§3). (2) An *identification rubric* of eleven auditable dimensions and a *flaw taxonomy* mapping each known threat one-to-one onto a dimension. (3) A *flaw-injection evaluation methodology* that yields reproducible local ground truth—detection, false alarm, and localization—without the true effect (§4). (4) A controlled comparison isolating evidence *presence* from evidence *adequacy*: once the leakage that makes naive evaluation circular is removed, a keyword baseline detects only 0.182 of injected flaws and is blind to commission-type threats, whereas swapping *only* the assessment policy to a reasoning model recovers detection to 1.000 with

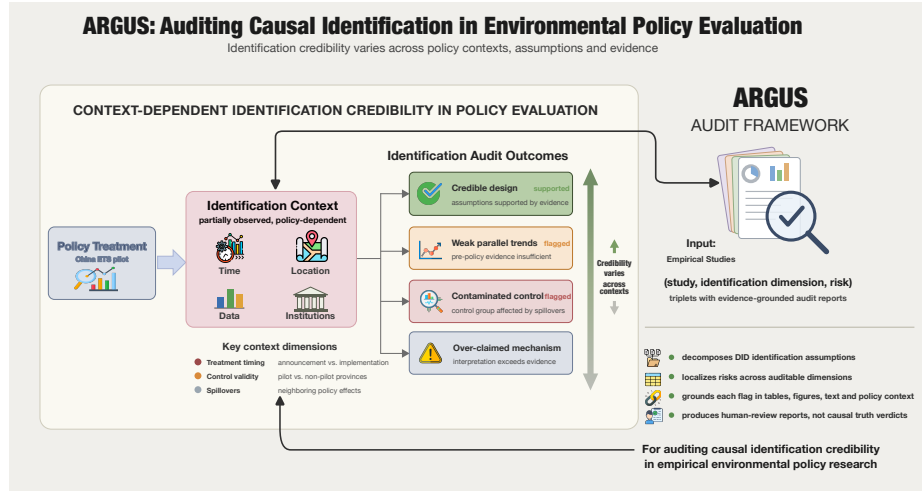


Fig. 1. Context-dependent identification credibility in empirical environmental policy evaluation. Causal-identification risks vary across treatment timing, control-group validity, spillovers, concurrent policies and other contexts, which ARGUS audits by decomposing assumptions and localizing evidence-based risk flags.

no false alarms (§5)—evidence that adequacy reasoning, not keyword matching, is what the task requires.

2 Related Work

DID identification and its threats. A large recent econometrics literature has formalized when DID estimators fail. Goodman-Bacon [4] decomposes the two-way fixed-effects estimator under variation in treatment timing and exposes the “forbidden” comparisons it makes between newly and already-treated units; de Chaisemartin and D’Haultfœuille [2] show such estimators can place negative weights on treatment effects; and Callaway and Sant’Anna [1] and Sun and Abraham [10] propose estimators robust to heterogeneous, staggered effects. Roth et al. [9] survey the design’s assumptions and the subtleties of pre-trend testing. ARGUS does not add an estimator; it *operationalizes* this literature as an auditable rubric, turning each known threat into a dimension whose supporting evidence can be checked in a given paper (§3).

Responsible AI for empirical research. A growing body of work applies AI to research reliability—automated reproducibility checking, statistical-reporting validation, and AI-assisted peer review. ARGUS is complementary but differs in target and in epistemic posture. First, it audits *identification credibility* specifically, the part of an empirical argument that determines whether a causal claim is licensed at all, rather than surface statistical reporting. Second, and centrally,

it does not issue verdicts: it produces evidence-anchored, human-reviewable reports that localize risk, leaving adjudication to a domain expert. This human-in-the-loop, “flag-not-judge” stance is a deliberate response to the fact that the counterfactual is unobserved and that automated causal verdicts would be both unreliable and unaccountable.

Benchmark and evaluation methodology. Our evaluation design follows three lessons from the benchmarking literature. Holistic evaluation [7] argues against collapsing capability to a single accuracy number; ARGUS likewise reports a layered profile—retrieval quality, relevance, adequacy, and localized risk—rather than one detection rate. Dynamic, adversarial benchmarking [6] warns that static benchmarks saturate; our flaw-injection loop is an adversarial, model-in-the-loop construction that we intend to escalate as the assessor improves (our synthetic flaws already saturate, motivating harder ones). Finally, Raji et al. [8] caution that turning an abstract capability into a measurable task is a consequential, non-neutral move; we take this seriously in §6, where we state precisely what ARGUS does and does not measure. Should the annotated corpus be released, we will document it with a datasheet [3].

Execution-grounded verification vs. assumption auditing. A complementary line asks whether a model can *carry out* causal inference correctly. CausalVerify [12], an execution-grounded benchmark, runs a model’s generated estimation code against synthetic data and checks that the recovered effect matches the canonical estimator on the realised dataset. ARGUS audits a different layer: not whether the estimation is *executed* correctly, but whether the identification *assumptions* behind it are adequately supported by the paper’s reported evidence—which CausalVerify explicitly places outside its scope (“identification judgment ... and assumption checking ... outside the direct scope”). ARGUS builds on the corpus of published economics papers released with that benchmark [13], adding a new dimension-level identification-risk annotation layer on top of it.

Bounded LLM agents. Tool-using agents in the ReAct style interleave reasoning and acting over an open horizon [11]. ARGUS deliberately *bounds* this paradigm. Agency is confined to two scoped sub-tasks (extraction and assessment), each run as a fixed-tool, hard-step-budget loop with a logged trace, while the inter-module control flow is ordinary deterministic code. We trade autonomy for two properties the framework depends on: *reproducibility*, so that the flaw-injection metrics are stable and comparable across runs, and *evidence-traceability*, so that every flag a human reviews is anchored to inspectable evidence. We view this restraint as a feature for responsible auditing rather than a limitation.

3 The ARGUS Framework

3.1 Identification rubric

ARGUS decomposes a DID study into eleven auditable dimensions: parallel trends, no anticipation, staggered-timing handling, SUTVA/spillovers, control-

group construction, specification, inference, sample-period selection, concurrent policies, placebo/robustness, and data measurement. Each dimension is expressed as an *assumption* (what must hold for identification), a testable *implication* (what should therefore be observable), and the *evidence* a credible paper would report to support it—for example, the parallel-trends dimension expects an event-study or pre-trend test with null pre-period coefficients. The rubric is declarative, stored in a configuration file (`identification_dimensions.yaml`), so that the set of audited dimensions is explicit and inspectable rather than buried in code. A companion flaw taxonomy (§4) maps each known identification threat one-to-one onto a dimension, which both documents the rubric’s coverage and supplies the perturbations used for evaluation.

3.2 Bounded-agentic pipeline

The audit runs a fixed, deterministic sequence—*decomposition* $\rightarrow$ *extraction* $\rightarrow$ *assessment* $\rightarrow$ *localization* $\rightarrow$ *report*—shown in the top band of Figure 2. The stage order is written in code and is never chosen by a model. *Decomposition* instantiates the rubric for the paper at hand. *Extraction* gathers, per dimension, the in-paper evidence relevant to that assumption. *Assessment* evaluates each assumption–implication–evidence chain and emits a per-dimension risk judgement with a rationale and cited evidence. *Localization* aggregates these judgements into a risk map that ranks where support is weakest. *Report* renders a transparent, evidence-anchored document for a human reviewer.

Agency is confined to the two stages where open-ended judgement actually helps, extraction and assessment. Each invokes a bounded ReAct-style loop with three constraints: (i) a *small fixed tool set* passed in per call—`evidence_search` over the paper’s text, figures and tables, `figure_parse` for structured figure/table content, and `policy_lookup` for policy metadata such as announcement versus implementation dates; (ii) a *hard step budget* (`max_steps`) so a run cannot wander indefinitely; and (iii) a *logged trace* of every step, tool call, and cited piece of evidence. All other stages are plain orchestration with no agent loop.

Why bound the agency. A fully autonomous agent would make per-run behaviour path-dependent and high-variance, which would undermine the central evaluation claim of §4—that flaw injection provides *reproducible* local ground truth. Restricting agency to two clearly scoped sub-tasks preserves reproducibility (stable, comparable metrics) and evidence-traceability (every judgement anchored to logged evidence a human can inspect). The expert sits at the end of the pipeline: ARGUS surfaces and localizes identification risk, it does not adjudicate it.

3.3 Output

ARGUS emits (*study*, *identification dimension*, *risk*) triplets, each with an evidence-grounded rationale, together with the human-readable report. It never produces a verdict on the true causal effect. This output shape is what makes the system both auditable by a human and measurable by the evaluation that follows.

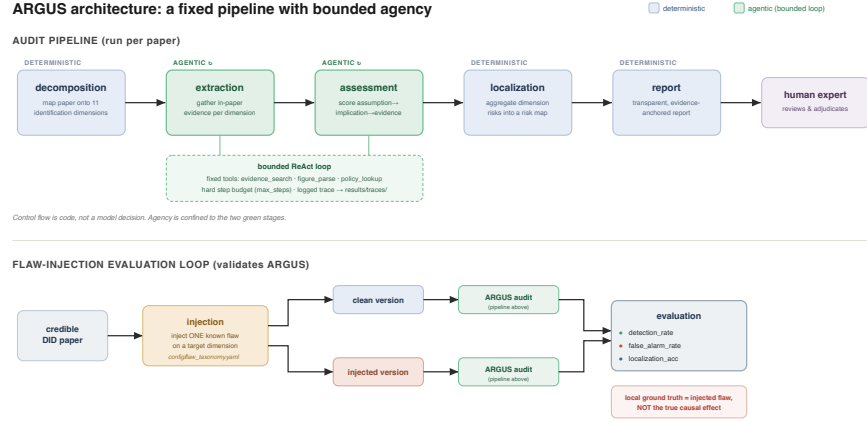


Fig. 2. ARGUS architecture. *Top:* the audit pipeline is a fixed deterministic sequence; agency is confined to the two green stages, each a bounded ReAct loop with a fixed tool set, a hard step budget, and a logged trace. *Bottom:* the flaw-injection evaluation loop injects one known flaw, audits the clean and injected versions, and scores detection, false alarm, and localization against the injected flaw as local ground truth—never against the unobserved true effect.

4 Evaluation by Flaw Injection

4.1 Local ground truth without the counterfactual

The defining difficulty in evaluating a causal-identification auditor is that the quantity of ultimate interest, the true causal effect, is never observed. We therefore do not attempt to certify that ARGUS recovers true effects. Instead we construct *local* ground truth by controlled perturbation. Beginning from a credible paper, we inject exactly one known flaw that targets a single identification dimension, drawn from the flaw taxonomy (`flaw_taxonomy.yaml`). Because we control which dimension we damaged, we know the correct audit outcome for that dimension—without ever needing the true effect. The bottom band of Figure 2 shows the loop: the clean and injected versions are each audited, and three metrics are computed.

- **Detection:** does the injected version raise risk on the targeted dimension above a threshold?
- **False alarm:** does the clean version stay quiet on that dimension?
- **Localization:** is the targeted dimension the one the auditor flags most strongly?

Table 1. Keyword baseline vs LLM evidence-adequacy assessor under realistic flaw injection (`clean_supported`, `max_steps` = 1, eleven flaws). The keyword baseline collapses once the sentinel leakage that made evaluation circular is removed; swapping only the assessment policy to a reasoning model recovers detection.

Assessor	detection	false alarm	localization
keyword baseline	0.182	0.000	0.182
LLM (gpt-4o)	1.000	0.000	0.909
<i>(circular, pre-fix)</i>	1.000	0.000	1.000

4.2 Realistic perturbation, not sentinels

How a flaw is injected is methodologically load-bearing. A naïve injector appends a sentence that *names* the flaw—for instance, “treated and control have non-zero pre-period coefficients with significant leads”. If the detector then searches for exactly those phrases, detection becomes a tautology: the injector plants the very string the detector greps for. In our initial implementation this produced a perfect-looking but meaningless detection rate of 1.000.

We replace this with *realistic structural perturbation* of two kinds. *Omission* flaws *remove* the section, figure, or table that supplies the support for a dimension—for example, deleting the placebo-robustness section or the balance table—modelling a paper that simply omits the check; an auditor can flag this non-circularly by noticing the *absence* of evidence. *Commission* flaws *rewrite* a section in flawed-but-natural prose, the way a real flawed paper would describe a mistaken design choice (e.g. pooling all province-years in a two-way fixed-effects regression under staggered timing), *without* containing any of the phrases the detector looks for. We enforce the latter with a regression test that fails if an injected paper contains the target dimension’s negative-signal phrases, so the circularity cannot silently return. Detection is thereby a genuine task rather than a string match, which is what makes the results in §5 meaningful.

5 Preliminary Results

We compare two assessment policies under the same flaw-injection protocol and the *same retrieved evidence* (extraction is identical), so the comparison isolates how each judges that evidence. The first is a deterministic, non-LLM **keyword baseline**: it scores each dimension from the presence or absence of supporting and risk-indicating vocabulary. The second is an **LLM assessor** that judges evidence *adequacy*—whether the reported evidence actually establishes the assumption. We run both over all eleven flaws on the `clean_supported` fixture, injecting one flaw per run. Table 1 reports the headline metrics; the per-flaw breakdown is in `experiments/phase1_pilot/`, and Figure 3 is generated from these runs.

The keyword baseline is blind to commission flaws. The baseline never raises a false alarm, and it correctly flags the two omission flaws—`measurement_break` and `spillover_contamination`—whose vocabulary does not appear elsewhere in the paper, detecting them by the *absence* of evidence. But beyond those, detection is only 0.182 (Figure 3, panel B). Two failure modes explain it: *cross-section vocabulary bleed*, where deleting the placebo section does not help because another section also mentions “placebo”; and, more fundamentally, blindness to *every* commission flaw, where flawed-but-plausible evidence is present and uses the expected topic vocabulary. Keyword presence is not evidence adequacy.

Reasoning over adequacy closes the gap. Swapping *only* the assessment policy—extraction is unchanged, so both see the same evidence—to an LLM that judges adequacy raises detection from 0.182 to 1.000 with no false alarms and 0.909 localization. The model reads the flawed-but-natural prose the keyword scorer cannot: for the injected anticipation flaw it returns *high / flawed* with the rationale that “outcomes began changing between the policy announcement and rollout,” even though no sentinel keyword is present. This is the empirical core of the project—evidence *presence* versus *adequacy*—and it validates the bounded-agentic design: agency placed exactly where reasoning is required.

Scope. These are synthetic fixtures with a small number of dimensions; 1.000 is a proof-of-concept on clearly injected flaws, not a claim of a solved task. The sharper question is whether the result survives on real published papers, to which we now turn.

5.1 From synthetic fixtures to real papers

We next audit 27 published DID papers from the CausalVerify corpus [13]—empirical economics broadly, with environmental policy as the motivating case. Here there is no injected ground truth; we report the raw per-dimension risk distribution of each assessor (Figure 4).

Two regimes, and a relocated bottleneck. The keyword baseline is credulous: 91% of its judgements are *low*—on real papers it sees topic vocabulary and declares the dimension supported, flagging almost nothing. The two-stage LLM is differentiated, but its most informative output is that $\sim 39\%$ of judgements are **unknown**: the relevance gate could not surface evidence for that dimension, so the system says so rather than over-flagging. Two regimes appear—dimensions whose discussion is diffuse or lives in figures the text retrieval cannot reach (parallel trends, no anticipation, placebo) are dominated by **unknown**, while dimensions where a section is retrieved (sample period, concurrent policies, SUTVA) are judged inadequate. The central finding is methodological: moving from synthetic fixtures to real papers is bottlenecked not by causal reasoning but by *evidence grounding*, which is exactly why the two-stage design reports retrieval quality and an explicit **unknown** state distinct from a substantive high-risk judgement (§6).

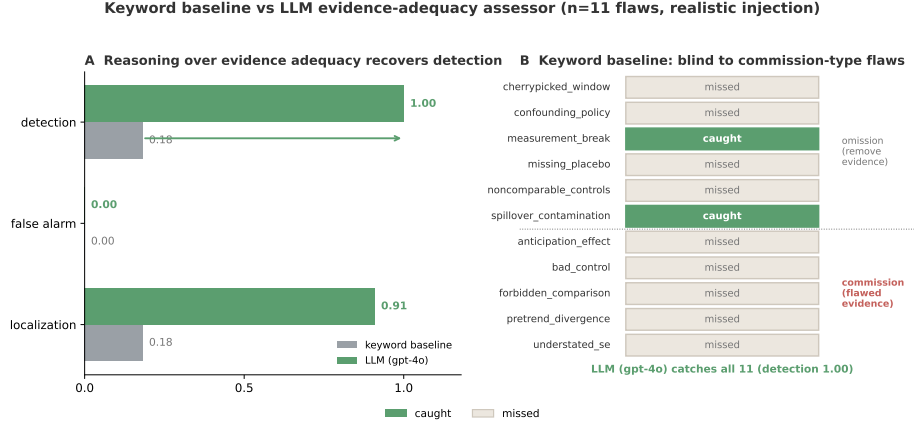


Fig. 3. Keyword baseline vs LLM evidence-adequacy assessor under realistic flaw injection. (A) Swapping only the assessment policy raises detection from 0.182 to 1.000 with false alarm fixed at 0 and localization 0.909. (B) Per-flaw keyword outcomes grouped by injection type: the baseline catches only omission-type threats whose vocabulary is not echoed elsewhere and is blind to every commission-type threat, all of which the LLM catches. Generated by `experiments/phase1_pilot/make_figure2.py` (keyword recomputed live; LLM headline frozen in `llm_summary.json`).

What the assessor actually reads. The judgements are grounded in the retrieved text, not pattern-matched. On a trade-and-wages paper ARGUS retrieves the employment-results section and flags *inference* as high because “the paper only mentions heteroskedastic-consistent standard errors and does not provide information on clustering or the number of clusters, which is crucial for the dependence structure in panel data”—a specific econometric observation, not a keyword hit. On another paper it retrieves an “Alternative Market Share Specification” section and flags *robustness/placebo* as high, noting that this “only discusses an alternative specification” and is not a placebo or falsification test. In both cases the model distinguishes a *related but different* reported check from the one the assumption actually requires—presence versus adequacy, now on real papers.

What we do not yet claim. Roughly 45% of the LLM’s judgements on these top-journal papers are *high*; without dimension-level expert labels we cannot say whether this is warranted or over-strict where retrieval succeeds. Calibrating ARGUS against expert annotation on this corpus is the immediate next step (§6); the defensible claim here is the contrast in behaviour and the localization of the grounding bottleneck, not the correctness of any individual flag.

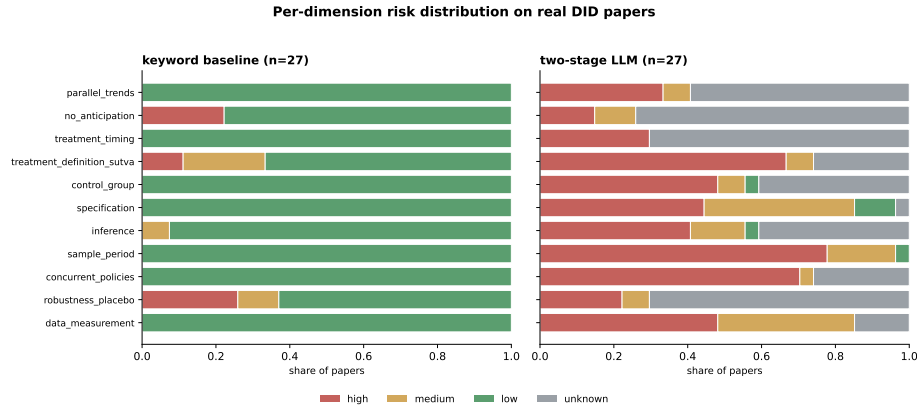


Fig. 4. Per-dimension risk distribution on 27 real DID papers (CausalVerify corpus). The keyword baseline (left) is almost entirely *low*; the two-stage LLM (right) is differentiated, with a large **unknown** share that honestly marks retrieval failure rather than over-flagging. Generated by `experiments/real_papers/make_realcorpus_figure.py` from committed per-paper CSVs.

6 Discussion and Research Plan

Threats to validity. The present results are deliberately preliminary. They use a small number of synthetic fixtures rather than a real corpus, and the baseline is an intentional lower bound, so its low detection rate should be read as a calibrated starting point, not as evidence about the eventual system. More fundamentally, flaw injection certifies an auditor against the *injected* threat on a single dimension; it does not certify the absence of other, un-injected threats, and it inherits whatever blind spots the flaw taxonomy has. We therefore treat detection and localization as necessary, not sufficient, evidence of an auditor’s quality, and pair them with the false-alarm rate to guard against a system that flags everything.

Construct validity: what ARGUS measures. A measurement is only as good as the link between the construct and the task that operationalizes it [5,8]. ARGUS audits whether a paper’s identification assumptions are *adequately supported by its reported evidence*—reported-evidence adequacy is an operational proxy for identification credibility, not causal validity itself. The two can diverge: a study may report every diagnostic yet still be invalid, or omit a check yet be sound. ARGUS therefore deliberately *flags rather than judges*, and a further gap separates the construct from the task—retrieving the relevant evidence. Our real-paper runs show this is the binding constraint: a reasoning assessor handed off-target evidence over-flags every dimension, so ARGUS reports a *retrieval-quality* status and an explicit **unknown** risk (“the system could not find the evidence”) distinct from a substantive high-risk judgement (“the paper lacks it”). The central methodological finding is thus that moving from synthetic fixtures to real

papers is bottlenecked not by causal reasoning but by *evidence grounding*, which is why ARGUS separates retrieval, relevance filtering, and adequacy assessment.

Research plan. The doctoral programme builds on the result in §5, in four stages. (1) *Harden the reasoning assessor.* The adequacy-reasoning policy already recovers detection on the synthetic fixtures; the next step is to stress it on harder, subtler flaws and to study its calibration, robustness across models, and the traceability of its cited evidence—under the same flaw-injection protocol, so gains stay attributable and reproducible. (2) *Dimension-level expert annotation.* We already audit a real corpus of published DID papers (§5.1); the missing piece is *dimension-level* expert labels of identification risk—the existing corpus is labelled at the method/claim level, not per assumption. Collecting these (with multiple annotators, recording disagreement) would calibrate whether ARGUS’s high/medium/**unknown** judgements match expert opinion and where experts themselves disagree. (3) *Human-expert studies.* Measure whether ARGUS reports actually help domain experts—calibration of trust, time saved, and errors avoided—since the system is explicitly designed to assist, not replace, expert judgement. (4) *Broader policy domains.* Extend beyond the China emissions-trading case to other environmental and, eventually, general empirical-economics settings, testing how far the rubric-plus-injection methodology generalizes.

7 Conclusion

We have argued that the right target for AI assistance in empirical policy research is not the unobservable causal effect but the *identification credibility* of the study that estimates it. ARGUS operationalizes this stance: a bounded-agentic pipeline decomposes identification into auditable dimensions, gathers and assesses in-paper evidence, and localizes risk in a transparent, human-reviewable report, while confining model agency to two scoped, traceable sub-tasks. We validate such auditors with synthetic flaw injection, which supplies reproducible local ground truth without access to the true effect, and we show that once the leakage making this evaluation circular is removed, a keyword baseline is blind to flawed-but-plausible evidence (0.182 detection) while a model that judges evidence adequacy recovers it (1.000, no false alarms)—evidence that presence is not adequacy. Scaling this from synthetic fixtures to a real, expert-annotated corpus with humans in the loop is the doctoral programme. We believe auditing identification credibility is both a tractable and a responsible role for AI in empirical environmental-policy research.

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